

第 3 問 複素解析: 調和関数

[1]

[1-1]

$$\begin{aligned} w &= \sin(x + iy) = \sin(x) \cos(iy) + \cos(x) \sin(iy) \\ u + iv &= \sin(x) \cosh(y) + i \cos(x) \sinh(y) \end{aligned} \quad (1.1)$$

[1-2]

$x = c$ を [1-1] に代入すると

$$u = \sin(c) \cosh(y), \quad v = \cos(c) \sinh(y) \quad (1.2)$$

となるので、

$$1 = \cosh^2(y) - \sinh^2(y) = \left(\frac{u}{\sin(c)} \right)^2 - \left(\frac{v}{\cos(c)} \right)^2 \quad (1.3)$$

これをプロットすると図 3.1 のようになる。

これよ領域 D_1 は、 w 平面上では $u \geq 0, v \geq 0$ に移されるのがわかる

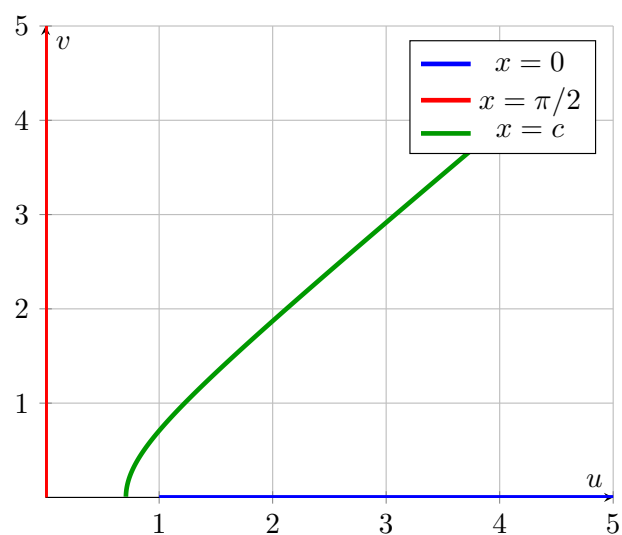


図 3.1: [1-2] の回答

[2]

[2-1]

Cauchy-Riemann の関係式より

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \quad (2.1)$$

これらの両辺を x 微分、 y 微分をして

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 v}{\partial x \partial y}, & \frac{\partial^2 u}{\partial x \partial y} &= -\frac{\partial^2 v}{\partial x^2} \\ \frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial^2 v}{\partial y^2}, & \frac{\partial^2 u}{\partial y^2} &= -\frac{\partial^2 v}{\partial x \partial y}\end{aligned}\tag{2.2}$$

これらを組み合わせると

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= 0 \\ \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} &= 0\end{aligned}\tag{2.3}$$

となり、示せた。

[2-2]

$$\begin{aligned}\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} &= \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \frac{\partial h}{\partial x} + \frac{\partial v}{\partial x} \frac{\partial h}{\partial v} \right) + \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial y} \frac{\partial h}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial h}{\partial v} \right) \\ &= \frac{\partial^2 u}{\partial x^2} \frac{\partial h}{\partial u} + \left(\frac{\partial u}{\partial x} \right)^2 \frac{\partial^2 h}{\partial u^2} + \frac{\partial^2 v}{\partial x^2} \frac{\partial h}{\partial v} + \left(\frac{\partial v}{\partial x} \right)^2 \frac{\partial^2 h}{\partial v^2} \\ &\quad + \frac{\partial^2 u}{\partial y^2} \frac{\partial h}{\partial u} + \left(\frac{\partial u}{\partial y} \right)^2 \frac{\partial^2 h}{\partial u^2} + \frac{\partial^2 v}{\partial y^2} \frac{\partial h}{\partial v} + \left(\frac{\partial v}{\partial y} \right)^2 \frac{\partial^2 h}{\partial v^2} \\ &= \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \frac{\partial h}{\partial u} + \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \frac{\partial h}{\partial v} \\ &\quad + \left\{ \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right\} \frac{\partial^2 h}{\partial u^2} + \left\{ \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 \right\} \frac{\partial^2 h}{\partial v^2} \\ &= \left\{ \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} \right\} \frac{\partial^2 h}{\partial u^2} \\ &= 0\end{aligned}\tag{2.4}$$

より、 $H(x, y)$ は調和関数

[3]

[3-1]

$z = \arcsin(w)$ より $0 \leq x \leq \frac{\pi}{2}, y \geq 0$.

(1) の条件のうち $u = 0$ は

$$0 = \sin(x) \cosh(y) \rightarrow x = 0\tag{3.1}$$

$v \geq 0$ は $\cos(x) \sinh(y) \geq 0$ となるため自動的に満たす。よって

$$H(0, y) = 0\tag{3.2}$$

(2) の条件のうち、 $v = 0$ は

$$0 = \cos(x) \sinh(y) \rightarrow x = \frac{\pi}{2} \quad \text{or} \quad y = 0\tag{3.3}$$

$u \geq 1$ は

$$\sin(x) \cosh(y) \geq 1\tag{3.4}$$

$x = \frac{\pi}{2}$ のときは $y \geq 0$, $y = 0$ のときは $x = \frac{\pi}{2}$ よりいずれにせよ $x = \frac{\pi}{2}$ が条件となる。よって

$$H\left(\frac{\pi}{2}, y\right) = 1\tag{3.5}$$

(3) の条件のうち、 $v = 0$ は

$$0 = \cos(x) \sinh(y) \rightarrow x = \frac{\pi}{2} \quad \text{or} \quad y = 0\tag{3.6}$$

$0 \leq u \leq 1$ の条件が $x = \frac{\pi}{2}$ のときは

$$\begin{aligned} 0 &\leq \sin(x) \cosh(y) \leq 1 \\ 0 &\leq \cosh(y) \leq 1 \\ y &= 0 \end{aligned} \tag{3.7}$$

$y = 0$ のときには

$$0 \leq \sin(x) \leq 1 \tag{3.8}$$

よってヤコビアンを計算して

$$\begin{aligned} 0 &= \frac{\partial x}{\partial v} \frac{\partial H}{\partial x} + \frac{\partial y}{\partial v} \frac{\partial H}{\partial y} \\ &= -\sin(x) \sinh(y) \frac{\partial H}{\partial x} + \cos(x) \cosh(y) \frac{\partial H}{\partial y} \end{aligned} \tag{3.9}$$

$x = \frac{\pi}{2}, y = 0$ のときは自動的に満たす。よって $y = 0$ のときを考えて、

$$\frac{\partial H}{\partial y} = 0 \tag{3.10}$$

まとめると

$$\begin{aligned} H(0, y) &= 0 \\ H\left(\frac{\pi}{2}, y\right) &= 1 \\ \frac{\partial H}{\partial y}(x, 0) &= 0 \end{aligned} \tag{3.11}$$

[3-2]

境界条件を眺めると H は y によらないことがわかる。なので、ラプラス方程式とその解は

$$\begin{aligned} \frac{dH}{dx} &= 0 \\ H &= Ax + B \end{aligned} \tag{3.12}$$

となる。

また、境界条件より

$$H = \frac{2}{\pi} x \tag{3.13}$$

[3-3]

$u = \sin(x)$ より、

$$h(u, 0) = \frac{2}{\pi} \arcsin(u) \tag{3.14}$$